

JOHN JERICK CAÑA AGAPITO

+63 969013037 | johnjerickagapito@gmail.com | Mohon, Iloilo City, Philippines
[linkedin.com/in/john-jerick-agapito-30651a337](https://www.linkedin.com/in/john-jerick-agapito-30651a337) | github.com/earlysev7n

SUMMARY

Computer Science undergraduate and capstone lead developer experienced in building cross-platform Flutter and Firebase applications with optimization algorithms, adaptive systems, and AI-assisted API integrations. Seeking a software engineering internship where I can contribute practical software development and problem-solving experience.

EDUCATIONAL BACKGROUND

Central Philippine University - Iloilo City, Philippines

2022 - 2027 (Expected)

Bachelor of Science in Computer Science

Relevant coursework: Data Structures and Algorithms, Software Engineering, Database Systems, Discrete Mathematics, Automata and Formal Languages, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Dart, Python, JavaScript, SQL

Frameworks & Libraries: Flutter, Provider, Node.js

Databases & Cloud: Firebase (Cloud Firestore, Authentication), MySQL

Tools: Git, GitHub, Visual Studio Code

AI & APIs: OpenAI API (GPT-4o-mini), REST API integration, structured JSON responses

Concepts: Genetic and greedy algorithms, adaptive systems, state management, data modeling, automated testing, agile development

PROJECTS

OneFit - Mobile Fitness and Nutrition Recommendation System

2026

Flutter, Dart, Firebase, Provider, REST APIs, OpenAI API | Capstone project, team of four | github.com/earlysev7n/onefit

- Led a four-person capstone team developing a cross-platform fitness app that generates personalized seven-day workout and meal plans from user goals, experience, equipment, schedule, and dietary restrictions.
- Engineered a constraint-aware greedy workout generator and a genetic meal optimizer using tournament selection, crossover, mutation, and elitism across 100 generations.
- Built a weekly adaptation engine that adjusts calorie targets by +/-100 kcal and workout difficulty using adherence, completion, effort ratings, and weight trends.
- Integrated ExerciseDB, USDA FoodData Central, OpenFoodFacts, and GPT-4o-mini for exercise data, nutrition search, barcode scanning, and structured recipe instructions.
- Designed an offline-capable Firebase architecture with user-scoped security rules and validated the algorithm layer with an extensive automated test suite.

CERTIFICATIONS

CCNA: Introduction to Networks — Cisco Networking Academy

2026

IT Essentials 8 — Cisco Networking Academy

2026

Introduction to Modern AI — Cisco Networking Academy

2026